Vehicle Parking Management System

Test Case Document

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UT = Unit Test, IT = Integration Test, ST = System Test. Test Case ID format: <type>-<FR number>_<case number>.

FR-01 — Register an account and log in using valid credentials.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-01_01	User Authentication	Valid registration	Registration page is available	Enter valid name, email and password; click Register	Name=Arjun; Email=arjun@example.com; Password=Abc@123	Account should be created and a confirmation shown.	Account was created and the confirmation was shown.	Pass
UT-01_02	User Authentication	Valid login	User account is registered	Enter registered email and password; click Login	Email=arjun@example.com; Password=Abc@123	User should be logged in and the home page shown.	User was logged in and a session was opened.	Pass
IT-01_03	User Authentication	Login with wrong password	User account is registered	Enter registered email and a wrong password; click Login	Email=arjun@example.com; Password=wrong-pass	System should reject login and display an error.	Login was rejected with "Invalid email or password".	Pass
IT-01_04	User Authentication	Reject duplicate email	Email is already registered	Enter an already registered email; click Register	Email=ARJUN@example.com	System should display "email already in use".	Registration was rejected with "Email is already in use".	Pass
IT-01_05	User Authentication	Reject invalid email format	Registration page is available	Enter a malformed email; click Register	Email=abc@com	System should display an email validation error and not create the account.	Validation error was shown and no account was created.	Pass
ST-01_06	User Authentication	Lock account after 5 failed logins	User account is registered	Enter a wrong password 5 times in a row	Email=arjun@example.com; wrong password x5	Account should be temporarily locked (SEC-04).	Not yet executed.	Pending

FR-02 — Register, view, update and manage vehicle information.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-02_01	Vehicle Management	Add a new vehicle	User is logged in	Open My Vehicles; enter vehicle details; click Save	Reg No=KA01AB1234; Type=Car	Vehicle should be saved and listed.	Not yet executed.	Pending
UT-02_02	Vehicle Management	View registered vehicles	User has registered vehicles	Open My Vehicles	N/A	List of the user's vehicles should be displayed.	Not yet executed.	Pending
IT-02_03	Vehicle Management	Reject duplicate vehicle	Vehicle is already registered	Enter the same registration number; click Save	Reg No=KA01AB1234	System should display "vehicle already exists".	Not yet executed.	Pending
IT-02_04	Vehicle Management	Reject invalid registration number	User is logged in	Enter an invalid registration number; click Save	Reg No=1234	System should display a validation error.	Not yet executed.	Pending
IT-02_05	Vehicle Management	Update vehicle details	Vehicle exists	Select vehicle; change colour; click Save	Colour=Red to Blue	Updated details should be saved.	Not yet executed.	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
ST-02_06	Vehicle Management	Updated vehicle persists after re-login	Vehicle exists	Update vehicle; log out; log in; open My Vehicles	Colour=Blue	Updated colour should be displayed.	Not yet executed.	Pending

FR-03 — View the current availability and status of parking slots.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-03_01	Parking Availability	Display slot status	Slots exist in the system	Open Slot Availability	N/A	Slots should show Available, Occupied, Reserved or Unavailable.	Not yet executed.	Pending
UT-03_02	Parking Availability	Filter available slots	Slots with mixed statuses exist	Open Slot Availability; apply Available filter	Filter=Available	Only available slots should be listed.	Not yet executed.	Pending
IT-03_03	Parking Availability	Availability after slot is occupied	A slot is assigned to a vehicle	Assign slot; refresh Slot Availability	Slot=P-12	Slot P-12 should show Occupied.	Not yet executed.	Pending
IT-03_04	Parking Availability	Availability after slot is released	Vehicle has exited	Process exit; refresh Slot Availability	Slot=P-12	Slot P-12 should show Available.	Not yet executed.	Pending
IT-03_05	Parking Availability	Attendant views availability	Attendant is logged in	Open Slot Availability as attendant	N/A	Same real-time slot data should be displayed.	Not yet executed.	Pending
ST-03_06	Parking Availability	Availability response time	Normal load conditions	Open Slot Availability; measure response time	N/A	Page should load within 2 seconds for 95% of requests (NFR-01).	Not yet executed.	Pending

FR-04 — Reserve an available parking slot for a specified period.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-04_01	Slot Reservation	Reserve an available slot	User is logged in; slot is available	Open slots; select slot; choose time period; click Confirm	Slot=P-05; 2 PM to 4 PM	Reservation should be created and the slot marked Reserved.	Not yet executed.	Pending
UT-04_02	Slot Reservation	Reservation requires login	User is logged out	Attempt to reserve a slot	Slot=P-05	User should be redirected to the login page.	Not yet executed.	Pending
IT-04_03	Slot Reservation	Reject already reserved slot	Slot is already reserved	Select the reserved slot; click Confirm	Slot=P-05 (reserved)	System should show an error and suggest another slot.	Not yet executed.	Pending
IT-04_04	Slot Reservation	Reject invalid time period	Slot is available	Select slot; enter end time before start time	Start=4 PM; End=2 PM	System should display a validation error.	Not yet executed.	Pending
IT-04_05	Slot Reservation	Reserved slot not given to another vehicle	Slot is reserved for 2 PM to 4 PM	Attempt entry into the reserved slot at 3 PM for a different vehicle	Slot=P-05; different vehicle	Assignment should be blocked (DR-08).	Not yet executed.	Pending
ST-04_06	Slot Reservation	Reservation shown in occupancy report	Reservation is created	Create reservation; open admin occupancy view	Slot=P-05	Occupancy count should include the reservation.	Not yet executed.	Pending

FR-05 — Record vehicle entry and create an active parking session.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-05_01	Vehicle Entry	Record vehicle entry	Attendant is logged in; slot is available	Enter vehicle number; click Confirm Entry	Reg No=KA01AB1234	Session should be created with the entry time recorded.	Not yet executed.	Pending
UT-05_02	Vehicle Entry	Reject unregistered vehicle	Vehicle is not registered	Enter unregistered vehicle number; click Confirm Entry	Reg No=KA99ZZ9999	System should display "vehicle not found".	Not yet executed.	Pending
IT-05_03	Vehicle Entry	Block second active session	Vehicle has an active session	Attempt entry for the same vehicle	Reg No=KA01AB1234 (active)	System should display "active session already exists" (DR-02).	Not yet executed.	Pending
IT-05_04	Vehicle Entry	Block entry when no slot is free	No slots are available	Attempt vehicle entry	Reg No=KA01AB1234; no free slot	System should display "no available slot".	Not yet executed.	Pending
IT-05_05	Vehicle Entry	Entry timestamp is correct	Attendant is logged in	Process entry; check recorded entry time	Reg No=KA01AB1234	Entry time should match the system clock.	Not yet executed.	Pending
ST-05_06	Vehicle Entry	Entry completes within time limit	Normal conditions	Perform entry; measure duration	N/A	Entry should complete within 3 seconds (NFR-02).	Not yet executed.	Pending

FR-06 — Assign an appropriate available slot and prevent duplicate assignments.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-06_01	Slot Assignment	Assign an available slot	Slot is available	Process entry; system assigns a slot	N/A	A suitable available slot should be assigned.	Not yet executed.	Pending
UT-06_02	Slot Assignment	Assign slot by vehicle type	Vehicle type is two-wheeler	Process entry for a two-wheeler	Type=Two-wheeler	A two-wheeler slot should be assigned (DR-07).	Not yet executed.	Pending
IT-06_03	Slot Assignment	Occupied slot not reassigned	Slot is already occupied	Attempt to assign the occupied slot	Slot=P-08 (occupied)	Assignment should be rejected (DR-03).	Not yet executed.	Pending
IT-06_04	Slot Assignment	Unavailable slot not assigned	Slot is marked unavailable	Attempt to assign the unavailable slot	Slot=P-15 (unavailable)	Assignment should be rejected.	Not yet executed.	Pending
IT-06_05	Slot Assignment	Attendant overrides slot	Several slots are available	Attendant selects a preferred slot; confirm	Slot=P-20	Selected slot should be assigned instead of the default.	Not yet executed.	Pending
ST-06_06	Slot Assignment	Slot shows Occupied everywhere	Slot is assigned	Assign slot; open Slot Availability	Slot=P-08	Slot should show Occupied on every screen.	Not yet executed.	Pending

FR-07 — Identify the current parking location and status of a parked vehicle.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-07_01	Vehicle Tracking	Find a parked vehicle	Vehicle has an active session	Enter vehicle number; click Search	Reg No=KA01AB1234	Current slot and status should be displayed.	Not yet executed.	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-07_02	Vehicle Tracking	Administrator tracks vehicle	Administrator is logged in	Search vehicle as administrator	Reg No=KA01AB1234	Vehicle location should be displayed.	Not yet executed.	Pending
IT-07_03	Vehicle Tracking	Vehicle not parked	Vehicle is not parked	Enter vehicle number; click Search	Reg No=KA09ZZ0000	System should display "no active session found".	Not yet executed.	Pending
IT-07_04	Vehicle Tracking	Empty vehicle number	Attendant is logged in	Leave vehicle number blank; click Search	Reg No=blank	System should display a validation message.	Not yet executed.	Pending
IT-07_05	Vehicle Tracking	Driver denied vehicle tracking	Driver is logged in	Attempt a staff-only action (track vehicle) as driver	Role=Driver	Access should be denied (SEC-02).	Driver was denied (403) on the staff-only route.	Pass
ST-07_06	Vehicle Tracking	Tracking shows latest slot	Vehicle was moved to a new slot	Reassign slot; search vehicle	Reg No=KA01AB1234	The new slot should be shown, not the old one.	Not yet executed.	Pending

FR-08 — Record vehicle exit and complete the parking session.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-08_01	Vehicle Exit	Record vehicle exit	Vehicle has an active session	Enter vehicle number; click Confirm Exit	Reg No=KA01AB1234	Exit time should be recorded and the session completed.	Not yet executed.	Pending
UT-08_02	Vehicle Exit	Slot released after exit	Exit is processed	Process exit; check slot status	Slot=P-08	Slot status should change to Available (SYS-12).	Not yet executed.	Pending
IT-08_03	Vehicle Exit	Block exit without active session	Vehicle has no active session	Attempt exit for a vehicle that is not parked	Reg No=KA09ZZ0000	System should display "no active session" (DR-04).	Not yet executed.	Pending
IT-08_04	Vehicle Exit	Reject exit before entry time	Active session exists	Set an exit time earlier than the entry time	Entry=5 PM; Exit=3 PM	System should display a validation error (DR-09).	Not yet executed.	Pending
IT-08_05	Vehicle Exit	Exit triggers fee calculation	Active session exists	Process exit; check the displayed fee	Entry=10:00; Exit=12:00	Fee should be calculated and shown before completion (UC-09).	Not yet executed.	Pending
ST-08_06	Vehicle Exit	Completed session cannot be edited	Session is completed	Attempt to edit the completed session	Session ID=S-1001	Edit should be blocked (DR-10).	Not yet executed.	Pending

FR-09 — Calculate parking duration and charges from the configured rates.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-09_01	Fee Calculation	Fee for whole hours	Entry and exit times are available	Process exit; system calculates fee	Entry=10:00; Exit=12:00; Rate=₹20/hr	Fee of ₹40 should be displayed.	Not yet executed.	Pending
UT-09_02	Fee Calculation	Fee for a partial hour	Entry and exit times are available	Process exit after a partial hour	Entry=10:00; Exit=10:45; Rate=₹20/hr	Fee should be rounded by the configured rule and displayed.	Not yet executed.	Pending
IT-09_03	Fee Calculation	Fee uses configured rate	Administrator has set the rate	Set rate; process exit	Rate=₹30/hr; Duration=1.5 hr	Fee of ₹45 should be displayed.	Not yet executed.	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
IT-09_04	Fee Calculation	Zero-duration stay	Entry and exit at the same time	Process exit immediately after entry	Entry=Exit time	Minimum fee should apply with no error.	Not yet executed.	Pending
IT-09_05	Fee Calculation	Rate change applies to new exits	New rate is configured	Update rate; process a new exit	New rate=₹25/hr	Fee should use the updated rate.	Not yet executed.	Pending
ST-09_06	Fee Calculation	Fee shown without delay	Normal load	Process exit; measure time to show fee	N/A	Fee should be shown within the 3-second exit flow (NFR-02).	Not yet executed.	Pending

FR-10 — Make and record parking payments.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-10_01	Payment Management	Successful payment	Fee is generated	Select payment method; click Pay	Amount=₹40; Method=UPI	Payment should be recorded and the session marked Paid.	Not yet executed.	Pending
UT-10_02	Payment Management	Payment options listed	Fee is generated	Open the payment screen	N/A	Available payment methods should be listed.	Not yet executed.	Pending
IT-10_03	Payment Management	Failed payment recorded	Payment gateway is available	Submit payment; gateway declines	Amount=₹40; invalid card	Status should be Failed and the user notified.	Not yet executed.	Pending
IT-10_04	Payment Management	Reject expired card	Payment gateway is available	Enter an expired card; click Pay	Expired card number	System should display a decline error.	Not yet executed.	Pending
IT-10_05	Payment Management	Attendant takes payment	Attendant is logged in	Attendant selects payment method; confirms payment	Amount=₹40; Method=Cash	Payment should be recorded against the session.	Not yet executed.	Pending
ST-10_06	Payment Management	Payment linked to session	Payment is completed	Complete payment; open the session record	Session ID=S-1001	Payment should be linked to the correct session (SYS-13).	Not yet executed.	Pending

FR-11 — Maintain and display historical parking-session information.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-11_01	Parking History	View completed sessions	User has past sessions	Open Parking History	User account with history	Past sessions with fee and status should be listed.	Not yet executed.	Pending
UT-11_02	Parking History	Session record fields	Completed session exists	Open Parking History; open a session record	Session ID=S-1001	Vehicle, slot, entry/exit, duration, fee and status should be shown.	Not yet executed.	Pending
IT-11_03	Parking History	Driver cannot edit history	Completed session exists	Attempt to edit a past session	Session ID=S-1001	Edit should be disabled or blocked (DR-10).	Not yet executed.	Pending
IT-11_04	Parking History	Empty history for new user	New user with no sessions	Open Parking History	New user account	A no-records message should be shown.	Not yet executed.	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
IT-11_05	Parking History	Administrator views a user's history	Administrator is logged in	Open a selected user's history	User=Driver X	That user's history should be displayed.	Not yet executed.	Pending
ST-11_06	Parking History	History with many sessions	User has 50+ sessions	Open Parking History; scroll or page through	User with 50+ sessions	All records should be reachable without errors.	Not yet executed.	Pending

FR-12 — Manage users, vehicles, slots and rates, and generate parking reports.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-12_01	Administration and Reports	Generate occupancy report	Administrator is logged in	Open Reports; select Occupancy; click Generate	Date range=Today	Report should show current occupancy.	Not yet executed.	Pending
UT-12_02	Administration and Reports	Add or remove a user	Administrator is logged in	Open User Management; add or remove a user	New user details	User list should update accordingly.	Not yet executed.	Pending
IT-12_03	Administration and Reports	Only administrators see audit logs	Attendant is logged in	Attempt to open audit logs as attendant	Role=Attendant	Access should be denied (SEC-08).	Not yet executed.	Pending
IT-12_04	Administration and Reports	Rate change reflects in fee	Administrator is logged in	Update rate; process a new exit	New rate=₹25/hr	Fee should use the updated rate.	Not yet executed.	Pending
IT-12_05	Administration and Reports	Disable a parking slot	Administrator is logged in	Disable a slot; open Slot Availability	Slot=P-30	Slot should show Unavailable and not be assignable.	Not yet executed.	Pending
ST-12_06	Administration and Reports	Audit log records admin actions	Administrator performs an action	Update a rate; open audit log	Action=rate update	Log should show the action, administrator and time (SEC-07).	Not yet executed.	Pending